Social Desirability in Large Language Model Responses: A Vignette-Experimental Study of Synthetic Respondents

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Abstract

This study examines how system-prompt instructions affect large language models (LLMs) used as synthetic respondents in four vignette experiments: coercive interrogation, charitable giving, energy conservation, and bribery. We compare synthetic responses with published human treatment contrasts and analyze final answers, verbalized reasoning, and exploratory hidden-state representations. Replication of human effects is uneven: some treatment directions agree, while others differ in sign or magnitude. System-prompt effects vary substantially across tasks and models. Direct anti-sanitization instructions can produce large shifts in responses and annotated argumentative content, whereas positive and negative framings of research consequences generally yield smaller and less consistent changes. Reasoning categories are associated with final answers, but direction-of-argument effects are less robust, and neither verbalized reasoning nor hidden-state differences establish a causal mechanism. Overall, steering answers away from socially approved positions does not reliably improve correspondence with human experimental effects. The findings underscore the importance of model identity, prompt design, and human benchmarks when using LLMs for experimental research.

The authors used AI to assist with code development, data analysis workflows, data markup and to improve the language and translation of the text. The authors reviewed and edited the output and take full responsibility for the content.

1. Introduction

Large language models (LLMs) are used to simulate survey respondents and pilot experimental designs [Argyle et al., 2023; Dillion et al., 2023; Horton et al., 2023]. Plausible answers, however, do not establish replication of human treatment effects. Socially sensitive vignettes sharpen this problem: human responses may reflect social desirability, while LLM outputs also depend on safety alignment and instructions [Krumpal, 2013; Salecha et al., 2024; Sharma et al., 2024; Tourangeau & Yan, 2007]. Moving an answer away from a socially approved position does not necessarily make it more human-like.

We study four adapted experiments: Hassner's coercive interrogation, Savary and Goldsmith's charitable giving, Bolsen et al.'s energy conservation, and Corbacho et al.'s corruption information. We compare synthetic treatment contrasts with published human benchmarks and estimate the effects of additional system prompts. Prompts are treated as experimental inputs, not validated corrections for social desirability bias.

Research question. How do system-prompt stimuli affect LLM answers, verbalized reasoning, and internal representations in socially sensitive vignette experiments?

Goal and objectives. We (i) compare original treatment contrasts with human benchmarks; (ii) estimate stimulus shifts and treatment-by-stimulus differences; (iii) analyze annotated arguments and their associations with answers; and (iv) explore stimulus and answer information in a separate local model's hidden states. These levels are distinct: textual associations and representational decodability do not establish causal mechanisms.

Behavioral responses and verbalized reasoning are analyzed across all four cases; hidden-state analysis concerns only the two initial cases and a separate local model.

2. Literature Review

2.1. LLMs as synthetic respondents

LLMs offer inexpensive ways to pilot surveys and experimental vignettes. Early work explored demographic conditioning and *algorithmic fidelity* [Argyle et al., 2023], classical experimental replication [Aher et al., 2023], and simulated economic agents [Horton et al., 2023]. Their usefulness depends not only on plausible answers but on reproducing human treatment contrasts. Replication studies find some directional agreement alongside exaggerated magnitudes, compressed variation, and weaker correspondence in socially sensitive domains [Bisbee et al., 2024; Cui et al., 2025; Röttger et al., 2024; Santurkar et al., 2023]. We therefore benchmark LLM treatment contrasts against published human studies rather than assuming that synthetic samples are representative.

2.2. Social desirability and instruction sensitivity

Social desirability bias (SDB) is response distortion associated with perceived acceptability. Survey theory distinguishes impression management from self-deceptive enhancement and emphasizes question sensitivity, privacy, and administration mode [Crowne & Marlowe, 1960; Krumpal, 2013; Paulhus, 1984; Tourangeau & Yan, 2007]. Indirect elicitation and anonymity can reduce some hu-

man response pressures [Horiuchi et al., 2022; Nederhof, 1985]; system prompts are not established equivalents of these survey methods.

LLM behavior introduces additional explanations. Models may favor socially approved answers, conform to perceived user expectations, or respond to safety-related training and instructions [Salecha et al., 2024; Sharma et al., 2024]. Survey prompts and response formats can also change their outputs [Tjuatja et al., 2024]. A shift away from a socially approved answer therefore need not represent reduced human-like SDB. This motivates studying system-prompt stimuli as experimentally consequential inputs and distinguishing behavioral steering from correspondence with human treatment effects.

2.3. Consequence framing versus gain–loss framing

Human studies establish that framing can affect decisions, but the magnitude varies substantially across problem designs [Kühberger, 1998]. Reviews of persuasive messages likewise do not find a general advantage of loss over gain framing [O'Keefe & Jensen, 2006]. Importantly, those studies manipulate how a choice or message is presented to human respondents. Our N−/N+ conditions instead frame consequences in *system-level instructions* to a synthetic respondent, rather than presenting matched gains and losses within the vignette. Their effects need not replicate classic gain–loss framing.

2.4. Reasoning and internal representations

Chain-of-thought prompting can alter generated outputs [Kojima et al., 2022; Wei et al., 2022], but verbalized reasoning is not a validated record of a model's internal causal process. We distinguish final answers, coded argumentative content, and hidden-state measurements. Annotation can identify associations between expressed arguments and responses, conditional on its reliability. Hidden-state probes offer a separate representational diagnostic; in our design, they concern a different local model and cannot be treated as a direct explanation of API-model behavior.

Research contribution

This study compares original treatment effects and system-prompt effects across four adapted experimental domains, with published human contrasts as benchmarks. It examines verbalized reasoning across all four cases and exploratory hidden-state representations for the two initial cases. The extension tests whether prompt sensitivity generalizes across models, normative contexts, and response formats without assuming that movement away from socially desirable answers improves human validity.

3. Research Design and Data

3.1. Case selection and experimental treatments

We study four published *attitude-expression* vignette experiments. Unlike autobiographical self-report questions, these tasks ask for judgments about externally described situations. This choice permits controlled variation of vignette treatments without assigning demographic identities or personal histories to the model. The original cases also differ in the normative direction of the response: endorsement of coercive interrogation is socially disapproved, whereas charitable giving is socially approved. We use this contrast to investigate whether system-prompt stimuli have similar or domain-specific effects.

Hassner: coercive interrogation. Following [Hassner, 2023], models evaluate the use of sleep deprivation in interrogating a terrorism suspect. The four conditions are a base case, modest expected intelligence value, protracted duration, and their combination. The response is binary (support versus opposition). Opposition is treated as the *normatively approved response direction* for descriptive comparisons, not as a verified measure of an individual respondent's latent attitude.

Savary and Goldsmith: charitable donation. Following [Savary & Goldsmith, 2020], models report willingness to donate $5 on a five-point scale. The three vignette conditions are private donation, public recognition, and public recognition combined with an additional social benefit. Greater stated willingness is the normatively approved response direction. The public-recognition treatment is also substantively important because it may alter the interpretation of donation

as altruism or reputational signaling.

Bolsen et al.: energy conservation. The third experiment adapts the energy-conservation framing design of [Bolsen et al., 2014]. Its nine treatment cells combine attribution of responsibility (government, individuals, or neither) with economic-cost, environmental-benefit, or no consequence framing. Models rate likelihood of investing in home insulation or weatherization on a seven-point scale. We examine original treatment-versus-control comparisons as well as attribution-by-consequence contrasts. The dependent variable measures stated willingness, not observed investment.

Corbacho et al.: bribery and institutional information. The fourth experiment adapts [Corbacho et al., 2016], comparing a control with information about corruption prevalence and an institutional-inefficiency placebo. Respondents indicate whether they would offer a bribe to avoid a fine, selecting True, False, or Prefer not to respond. These are three nominal outcomes; refusal is not interchangeable with unwillingness. The source paper reports direct-response and privacy-adjusted comparisons with different estimands. Our prompt adaptation does not reproduce its original crosswise-response procedure.

Because original elicitation procedures and outcomes differ, we compare treatment directions without pooling numerical effect sizes. Matching a direction does not establish matching response distributions or mechanisms.

3.2. Prompt structure and unit of observation

The original vignette treatment enters through the user-facing question and response options. The system message specifies the survey-response task; experimental system-prompt stimuli are added to this instruction layer. No demographic persona is assigned in the principal behavioral design. Stimulus wording is identified from available generation configurations; unverified historical H–M wording is not quoted.

Each run generates an answer and, where available, a linked reasoning trace. Repeated completions in the same model–experiment–treatment–stimulus cell measure within-cell variation, not independent human respondents; inference must respect that grouping.

3.3. Models, decoding, and analytical samples

The behavioral corpus uses four API-accessed reasoning-capable models: `moonshotai/kimi-k2-thinking`, `z-ai/glm-4.6`, `qwen/qwen3.5-9b`, and `minimax/minimax-m2.5`. These models were selected pragmatically for the availability of final answers and reasoning traces; they do not constitute a representative sample of all LLMs. A separate local `Qwen3-8B` is used for hidden-state analysis and is not pooled with the API behavioral sample.

The principal generation regime uses temperature $T = 0.01$, with a separate no-stimulus comparison at $T = 1.0$. Low temperature improves repeatability but may compress within-cell variation; the temperature comparison is a decoding sensitivity check rather than an independent human benchmark.

The historical corpus reports 15,680 raw runs, 12,434 retained rows, and 11,040 main $T = 0.01$ observations. Historical totals follow the available reconciled master-table copy; the H–M family was collected only for GLM and Kimi.

4. System-Prompt Stimuli

4.1. Conceptual role

System-prompt stimuli modify the instructions surrounding a fixed vignette. They are intended to discourage normatively polished or sanitized answers, but their observed effects are described here as *stimulus-induced response shifts*, not as established reductions in social desirability bias. The direction of a shift is evaluated relative to the *normatively approved response direction* in each case. A shift away from that direction does not, on its own, establish greater correspondence with human responses.

The historically implemented stimulus set includes the no-stimulus baseline, two core conditions (A and C), and the extended H–M family. A emphasizes the costs of sanitized answers; C includes an explicit anti-overcorrection qualification. The H–M family varies the stated target of consequences or benefits (science versus a company; personal or institutional benefit versus public good) and the valence of the framing (negative consequences versus positive benefits). These labels describe intended prompt functions, not independently identified causal mechanisms.

4.2. Historical stimulus conditions and coverage

Table 1: Historical stimulus families and interpretation

Condition	Interpretation and limitation
No stimulus	Baseline survey task without an additional experimental system-prompt stimulus.
A	Strong anti-sanitization / realism instruction invoking negative consequences for research quality; exact wording to verify.
C	Core realism instruction with an anti-overcorrection qualification; not a neutral control matched to A on all dimensions.
H–K	Negative consequence framing with different stated recipients or beneficiaries; fielded in a subset of models.
L–M	Positive benefit framing with different stated recipients or beneficiaries; fielded in a subset of models.

The existing coverage matrix contains H–M records for *both* Hassner and Savary. However, the available project reconstruction provides semantic descriptions rather than verified verbatim H–M prompt strings. We therefore avoid verbatim claims about H–M wording and do not treat this family as an orthogonal factorial design.

N0, N−, and N+ are also used in the new experiments and the Hassner/Savary extension; N0 is distinct from historical no stimulus.

4.3. Implemented stimulus conditions in the new experiments

Bolsen and Corbacho each use A, C, N0, N−, and N+ prompts. N0 explicitly instructs the model to respond with neutral realism; it is *not* the historical no-incentive condition. Similarly labeled prompts should be compared using the actual message text rather than their short identifiers alone. Therefore the new N0-referenced comparisons and the older unstimulated-baseline comparisons have different reference conditions, and are not treated as a fully matched four-experiment factorial experiment.

4.4. Analytical contrasts

The behavioral analysis distinguishes three comparisons: (i) original vignette-treatment contrasts within a fixed prompt condition; (ii) stimulus-

induced shifts relative to a no-stimulus or explicitly specified reference condition; and (iii) changes in original treatment contrasts across stimulus conditions. Attribute-level comparisons are interpreted as conditional associations unless the compared prompts isolate the attribute of interest and have sufficient balanced coverage. Reasoning annotations and hidden-state measures are examined as complementary outcomes, with their separate validity and generalizability limitations stated in the relevant methods sections.

5. Analytical Strategy

The unit is one model generation within an experiment–treatment–stimulus cell. Repeated low-temperature completions are not independent human respondents; all uncertainty estimates are interpreted conditionally on the generation and sampling protocols.

5.1. Behavioral outcomes

Hassner endorsement is binary (logit and penalized-logit sensitivity); Savary uses a five-point willingness score (OLS, ordinal, and threshold sensitivities); Bolsen uses a seven-point investment scale (cell means and ordinal distributions). Corbacho has three nominal outcomes—True, False, and refusal—which are never collapsed into an ordered score. We estimate original vignette-treatment contrasts within stimulus and model, stimulus shifts within original treatments, and their interactions. The principal reference is historical no stimulus, fresh `none`, or new neutral realism N0, as appropriate; these are distinct conditions.

Pairwise Holm corrections and broader Benjamini–Hochberg adjustments address multiplicity. Historical models use heteroskedasticity-robust and design-cell-clustered checks. Low-temperature repetitions, quasi-separation and saturated cells constrain conventional uncertainty estimates. The new-experiment analysis uses model-specific cell contrasts, ordinal/nominal distribution checks and conditional block resampling; the latter does not yield uncertainty for a human population. Historical H–M prompt-attribute regressions are exploratory because the observed prompt packages do not isolate framing valence or beneficiary in a full factorial. Fresh `none` versus N-series comparisons may also reflect differences in run timing and provider routing.

5.2. Verbalized reasoning

The historical annotation pipeline retained 8,638 records; its categories encode three-level argument centrality, signed direction where defined, and style. A 27-trace expert-coded subset shows uneven agreement. The new Bolsen/Corbacho batch contains 1,398 eligible traces over selected original treatments, and instead encodes binary category presence with signed direction where applicable. These schemes are not pooled as if they measured identical centrality. We estimate stimulus-related changes in category prevalence and, separately, argument direction among occurrences; undefined direction is not assigned zero. Within-cell regressions relate category presence to final outcomes with model, treatment, stimulus and version controls. Associations between co-generated traces and responses are not causal mediation. For rare and near-constant categories, estimates can be unidentified or imprecise.

5.3. Local hidden-state exploration

A separate Qwen3-8B study records 37-layer, 4,096-dimensional last-token representations for 98 runs under a different prompt inventory. Layer-wise distances, centered kernel alignment, matched trajectory contrasts, and predictive probes are descriptive diagnostics, not direct evidence of a mechanism in the API models. The study distinguishes observable shifts, recovery of published human treatment effects, and broader human validity throughout.

6. Results

6.1. Final answers: heterogeneous models and systematic prompt shifts

The four API models differ substantially in observed answers. Across the reported $T = 0.01$ analytical sample, Hassner endorsement rates range from 10.3% (Minimax) to 67.5% (GLM-4.6), while Savary mean willingness ranges from 3.33 (Kimi) to 3.73 (Qwen). The Hassner treatment-by-model specification also improves model fit (McFadden pseudo-R^2: 0.507 to 0.543; $\Delta\text{AIC} = -304$). Model identity therefore matters for both response levels and sensitivity to vignette manipulations. These descriptive rates pool different available stimulus conditions and should not be read as balanced model comparisons.

In the no-stimulus Hassner condition, support for sleep deprivation decreases from 44.6% in the control to 19.3% with protracted duration, 12.2% with modest intelligence value, and 1.0% with both modifications. These directions agree with the human experimental benchmark. In Savary, by contrast, the LLM mean rises from 3.39 in the private condition to 3.73 with public recognition and 4.09 with added social benefit. The increment associated with social benefit is directionally consistent with the source experiment, but the private-to-public increase reverses the human finding: public recognition *reduced* willingness in the original study. The benchmark therefore reveals domain-specific limits of directional replication rather than a general success or failure of synthetic respondents.

All eight historical system-prompt stimuli are associated with shifts *away from the normatively approved response direction* in the pooled descriptive comparisons. In Hassner, the no-stimulus endorsement rate is 19.1%; it is 41.6% under C, 72.1% under A, and ranges from 56.4% to 90.0% across H–M. In Savary, mean willingness is 3.74 without a stimulus, 3.44 under C, 3.32 under A, and 3.14–3.35 across H–M (Table 5). These are *socially desirable response patterns* and *stimulus-induced response shifts*, not direct estimates of human-style SDB or its mitigation.

The comparisons require a coverage qualification: H–M were fielded only on Kimi and GLM-4.6, while the baseline and core conditions have broader model coverage. Thus the displayed pooled means are useful descriptive summaries, but differences involving H–M are not automatically like-for-like causal estimates. In the historical attribute regression restricted to the two-model subsample, positive-framed L/M prompts are associated with substantially higher Hassner endorsement than negative-framed H–K (reported OR ≈ 15, $p < 0.001$) and a smaller decrease in Savary willingness (reported $\beta = -0.13$, $p = 0.02$). Because the observed prompts do not form a complete matched factorial, these are associations between prompt packages, not isolated effects of framing valence. Nonsignificant recipient and beneficiary coefficients do not establish that those attributes are inert.

Treatment-by-stimulus contrasts also vary: the original analysis reports moderation in some conditions, without reversal of the sign of the original Hassner treatment contrasts. In Savary, the private-to-public contrast remains positive across the eight historical stimuli (reported range $+0.05$ to $+0.26$),

so none reproduces the human reversal under the tested settings. This result identifies a limitation of the *observed prompts*, not a general impossibility of eliciting self-signaling-related reasoning.

Historical-data reconciliation. Reapplying Notebook 05 preprocessing to the available master table yields 11,040 observations in 168 model–treatment–stimulus cells; 16 cell estimates differ from an earlier export. The pooled baseline means above are consistent with the reconstruction. Model-specific no-stimulus Hassner modest-intelligence contrasts are -0.785 (GLM) and -0.442 (Kimi); Savary public-minus-private contrasts are $+0.238$ and $+0.400$. Directions remain unchanged. The exact provenance of the notebook's original input and some run-temperature metadata remains unresolved.

6.2. Fresh `none` replication: Hassner and Savary

A separate low-temperature October 2026 run without an additional incentive section produced 698 valid answers from 700 requests. In Hassner, GLM control endorsement is 1.000, falling to 0.673 (duration), 0.360 (modest intelligence), and 0.040 (both); Kimi's corresponding rates are 0.571, 0.140, 0, and 0. In Savary, private/public/social-benefit means are 3.48/3.92/4.42 (GLM) and 3.34/3.66/4.00 (Kimi). Both studies preserve the historical treatment-effect directions, including Savary's public-recognition reversal relative to humans. The new `none` sample is not pooled with historical runs; its contrasts against later N-series runs are descriptive because inference-provider routing and execution time differed.

6.3. N-series behavioral extension: Hassner and Savary

The N0/N$-$/N$+$ extension yields 2,065 valid answers from 2,100 requests (Kimi 1,043; GLM 1,022). In Savary, N0 relative to the separate `none` run lowers mean willingness by 0.15 points for Kimi (95% CI $[-0.23, -0.06]$, Holm $p = 0.013$) and by 0.12 for GLM (Holm $p = 0.339$). Compared with N0, both additional framings raise Kimi's mean by about 0.09, but lower GLM's by 0.14 (N$-$) and 0.21 (N$+$). The GLM N$+$ shift is -0.213 ($[-0.320, -0.107]$, Holm $p = 0.00117$); the corresponding GLM–Kimi difference in shifts is -0.307 ($[-0.447, -0.167]$).

In Hassner, mean N+–N0 changes are -6 percentage points (pp) for Kimi (95% CI $[-11.65, 0.18]$; Holm $p = 0.335$) and $+4$ pp for GLM (Holm $p = 1$). However, Kimi's control endorsement falls from 72% to 46%, an exploratory -26 pp local shift that changes treatment contrasts. Twenty-seven transport failures left only 23 of 50 valid GLM duration responses under N−. The Kimi mean contrast attenuates to about -2 pp after provider adjustment. Thus near-symmetric framings do not yield a universal effect, and cross-run contrasts cannot be assigned solely to prompt wording.

6.4. Additional experiments: Bolsen and Corbacho

Sample and comparability. The two new experiments contain 6,034 saved generations, 6,000 of which satisfy parsing and scale checks. GLM-4.6 and Kimi-K2-Thinking cover the full treatment-by-stimulus design; Qwen3.5-9B contributes only 33 valid Bolsen control/A responses and is excluded from the principal comparisons. The realized prompt conditions are A, C, N0, N−, and N+. N0 is a neutral realism instruction, not the historical no-incentive baseline; these references must not be equated.

Bolsen: partial directional replication and model heterogeneity. Under N0, both models' control means are 5.000 on the seven-point investment scale. Government attribution lowers investment likelihood by 0.840 points for GLM and 1.800 for Kimi, compared with approximately -0.370 in the published human contrast. Cost framing lowers it by 1.060 and 0.600 points, compared with a published contrast of approximately -0.360. The combined individual-attribution and environmental-benefit condition is only 0.020 points above control for either model, versus approximately $+0.450$ in the human study. Agreement in sign thus coexists with sizeable discrepancies in magnitude (Table 3). Published human contrasts are reconstructed from rounded means, and exact comparability is limited.

The factorial pattern is also heterogeneous. At N0, the individual-by-environment interaction is 0.000 for GLM and -0.180 for Kimi, against approximately $+0.120$ in the reported human means. Averaging equally across the nine treatments, A changes GLM's mean by $+0.168$ relative to N0 but Kimi's by -0.665; the corresponding C shifts are $+0.257$ and -0.097. The prompt-induced shift therefore has opposite signs across models, including a

rise in pro-conservation responses for GLM. In Kimi, the cost-minus-control contrast changes from -0.600 under N0 to $+0.080$ under A. These shifts neither guarantee greater human correspondence nor support a uniform de-sanitization claim.

Figure 1.

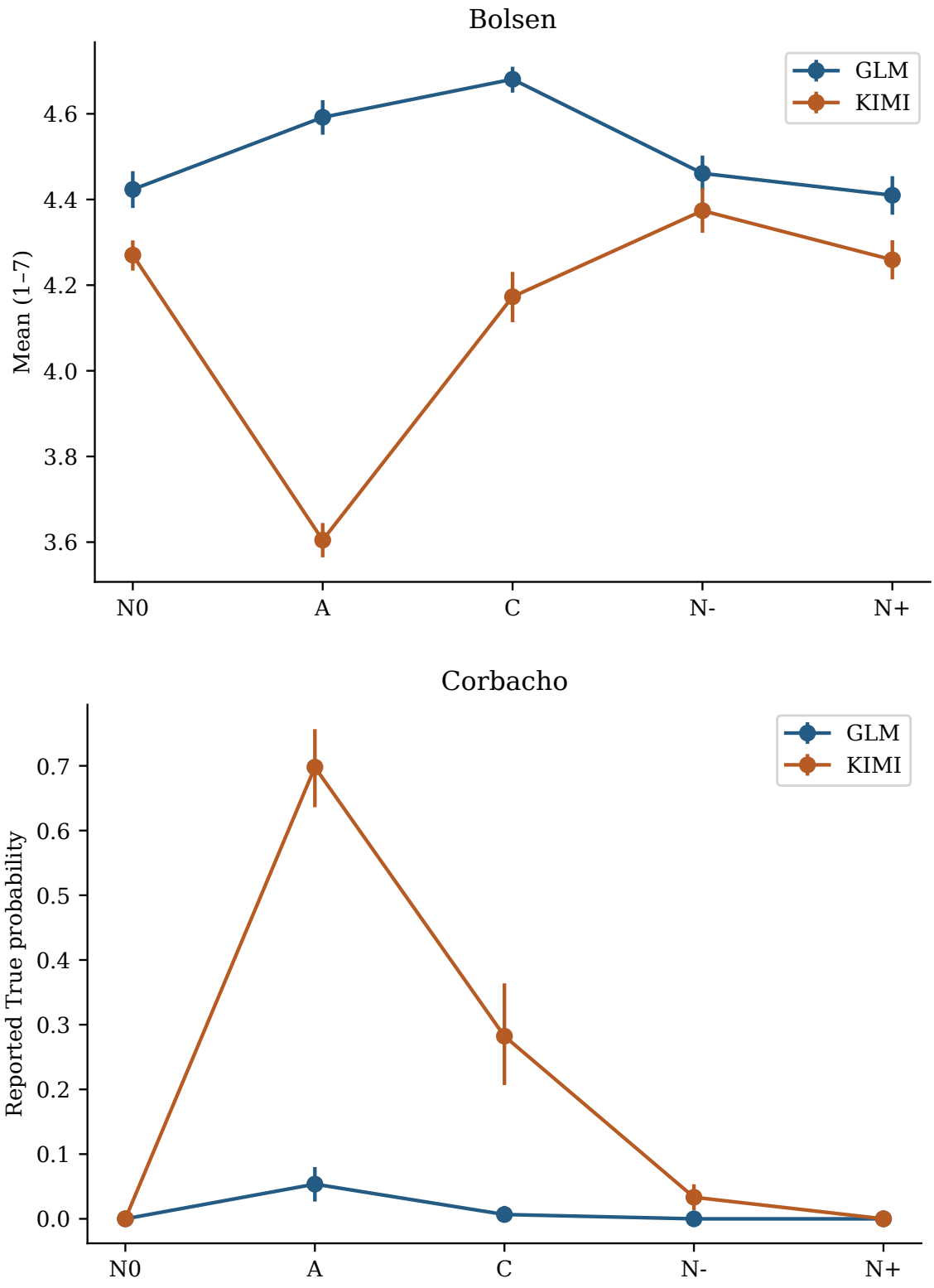


Figure 1: Treatment-balanced response levels by model and prompt. Equal treatment weights; pointwise block-10 conditional reference intervals. N0 is neutral realism. Bolsen: mean on the 1–7 scale; Corbacho: True probability.

Corbacho: three response outcomes and placebo sensitivity. Under N0, every valid GLM and Kimi answer in all three treatment groups is *False*; the observed treatment differences are zero. The closest-format published human

direct-response contrast for corruption exposure is approximately $+0.030$, with a reported confidence interval reaching approximately zero. A larger published privacy-adjusted joint contrast ($+0.080$) measures a different quantity and cannot be substituted. Human and synthetic elicitation procedures differ.

For Kimi under A, the proportions choosing True are 0.560 in control, 0.694 under corruption information, and 0.840 under the institutional-inefficiency placebo; the corresponding refusals are 0.380, 0.306, and 0.160. Under C, True proportions are 0.100, 0.300, and 0.447, with refusals of 0.440, 0.220, and 0.234. Thus the corruption-minus-placebo True contrast is approximately -0.146 under A and -0.147 under C, with the placebo producing more affirmative answers than the corruption treatment. The treatment-balanced True proportion under A is 0.054 for GLM but 0.698 for Kimi; no GLM refusals appear under A. Refusals are retained as a separate nominal response outcome, not coded as opposition or as an ordinal midpoint (Table 4).

Figure 2.

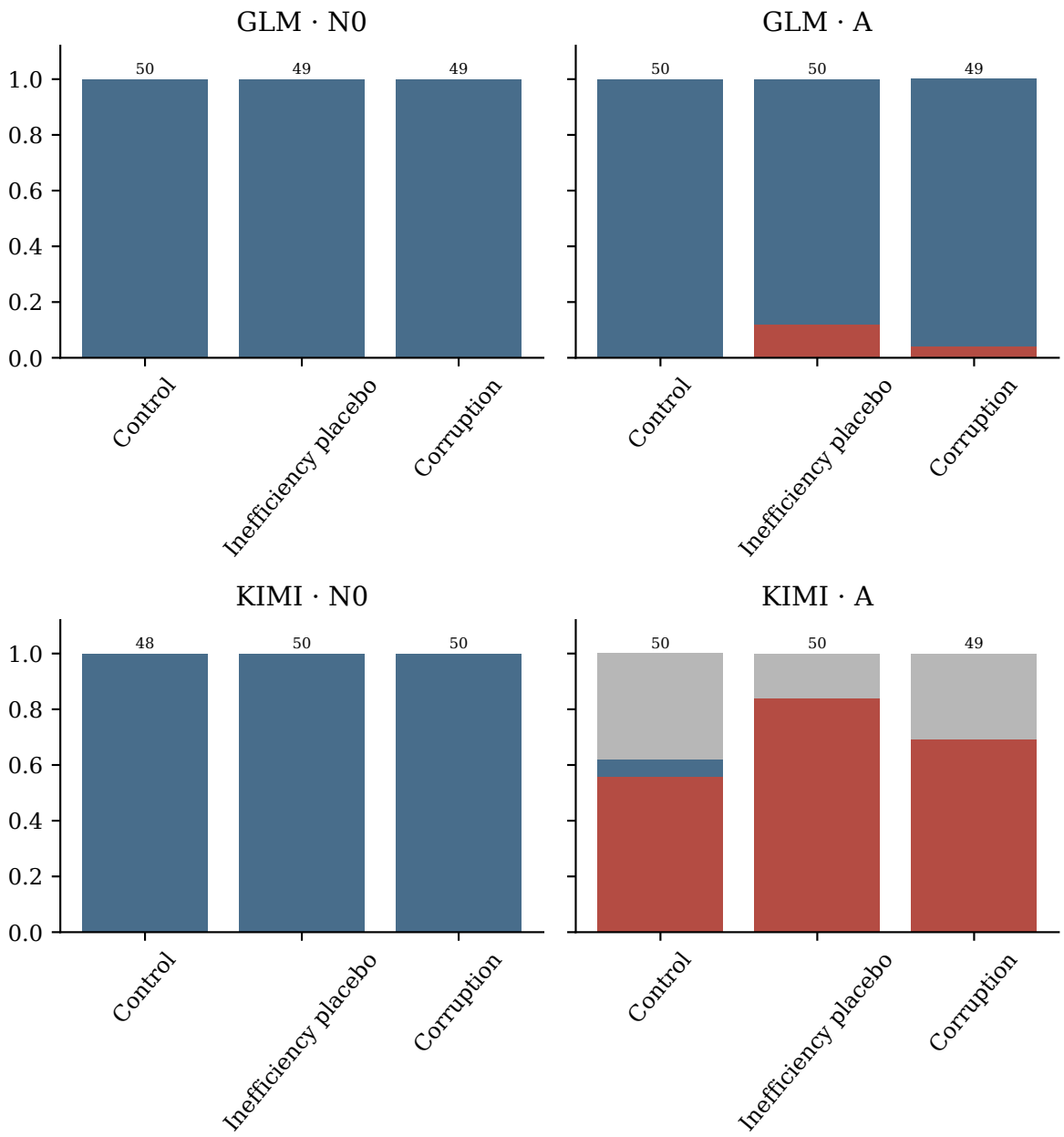


Figure 2: Corbacho response shares by treatment and model: N0 and A. Red: True; blue: False; gray: refusal. Labels show valid n; N0 is neutral realism. All three outcomes remain distinct.

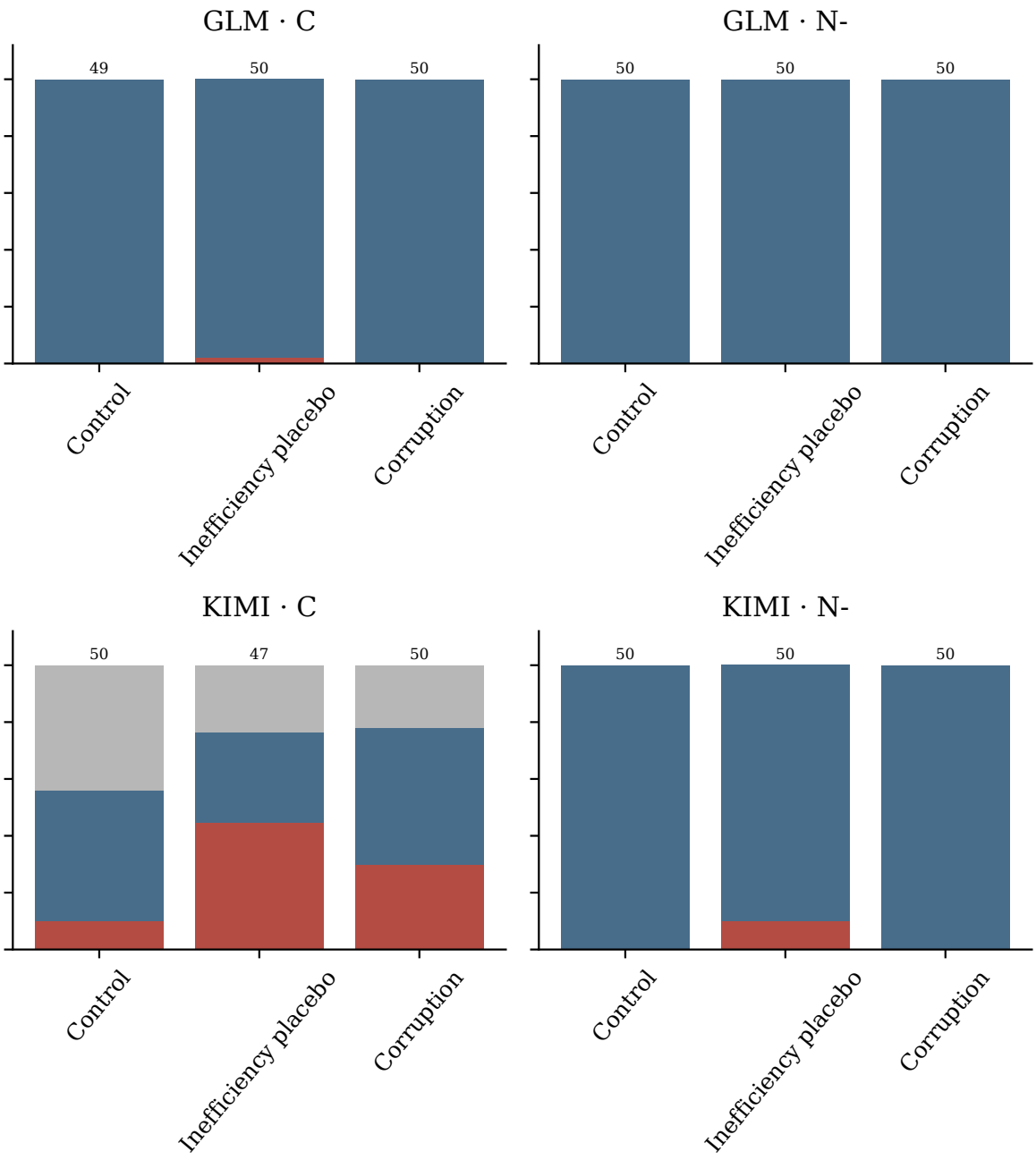


Figure 2: Corbacho response shares, continued: C and N−. Same outcomes and colors; vertical scale 0–1.

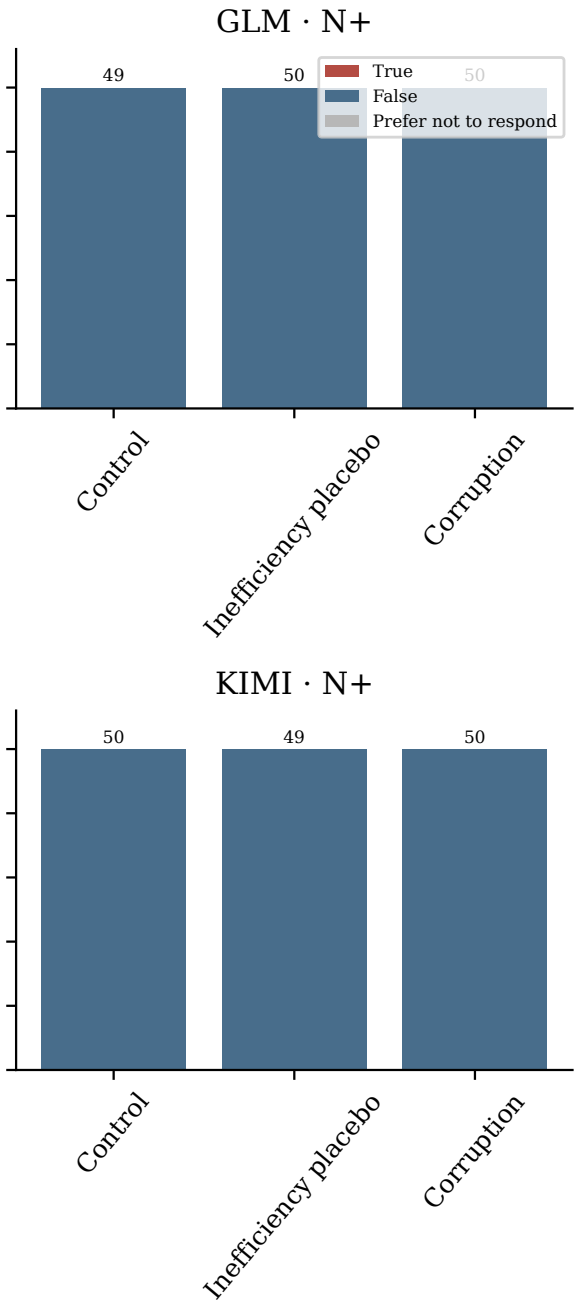


Figure 2: Corbacho response shares, continued: N+. Same outcomes and colors; vertical scale 0–1.

The new negative- and positive-consequence stimuli. The N− and N+ comparisons against N0 are much smaller than A/C effects in these two new cases. In Bolsen, treatment-balanced mean shifts under N− are $+0.038$ (GLM) and $+0.104$ (Kimi) scale points, compared with -0.014 and -0.011 under N+. Thus the N+ minus N− contrasts are -0.051 and -0.115, respectively. Under conditional block-of-ten resampling, the latter contrast has a reference interval of $[-0.186, -0.045]$ for Kimi but $[-0.112, 0.008]$ for GLM; these are not human-

population confidence intervals. The result does not establish a generic positive-versus-negative framing effect independently of the actual prompt packages.

In Corbacho, GLM has no observed True or refusal responses under either N− or N+ in the balanced comparisons. Kimi's True share changes by $+0.033$ (3.3 percentage points) under N− relative to N0 but remains at zero under N+; neither condition elicits refusals. Hence N+ minus N− in Kimi is -0.033 for True, with a conditional block-of-ten reference interval $[-0.053, -0.013]$. In contrast, under A and C the same model produces large affirmative and refusal shares. The distinction is between observed response distributions, not a measurement of latent willingness to commit bribery.

Four-case interpretation. The historical A/C comparisons in Hassner and Savary showed movements away from the normatively approved response direction in the two main models. Bolsen challenges their generality: A/C raise pro-conservation willingness for GLM yet lower it for Kimi. Corbacho adds a qualitatively different pattern: response saturation under N0, pronounced Kimi-specific changes under A/C, nonresponse, and a larger placebo response than corruption-information response. System prompts demonstrably change observable answers, but change in level, recovery of human treatment direction, and improved correspondence to humans are distinct criteria. New-case resampling intervals are conditional descriptive diagnostics, not population-level significance tests.

6.5. Verbalized reasoning: changes in content and argumentative direction

The annotation results indicate that stimuli change the text models produce while selecting answers. Explicit `overcorrection_concern` appears in approximately 9% of no-stimulus traces and 72–76% under A/C (expert-validation $\kappa = 0.64$, $n = 27$). The prevalence of explicitly rational-deliberative style rises from 0.67 to 0.86, and acknowledgement of multiple defensible positions from 0.57 to 0.80. The latter style codes were not included in the small human-validation set and should be treated as exploratory. These changes describe generated text, not independently verified cognitive awareness.

The A–C comparison is especially informative in Hassner. Endorsement is 72.1% under A versus 41.6% under C, while several coded category-prevalence

differences are small. Direction codes, however, differ markedly: signed presence of `ratio_of_goals_and_means` changes from -0.01 under C to $+0.42$ under A, `majority_reliance` from $+0.06$ to $+0.41$, and `cruelty` from -0.68 to -0.34 (Table 6). The first of these codes was not covered by expert validation; the others have limited or category-specific reliability evidence. The pattern is consistent with a shift in the *polarity of verbalized arguments*, but it does not show that these arguments mediate the stimulus effect.

In Savary, both A and C strongly increase annotation of personal costs and benefits, while their mean final answers remain comparatively close (3.32 and 3.44). Across both domains, the annotation vocabulary therefore responds to prompts, but the relation between coded text and the answer is domain-dependent.

A separability benchmark classifies stimulus-present versus no-stimulus traces from annotation features at reported cross-validated ROC-AUC 0.96, compared with 0.68 for low- versus high-temperature no-stimulus traces. The original analysis also reports that the contrast survives exclusion of directly stimulus-linked codes. This demonstrates that the two classification tasks have different separability under the chosen features and samples; it does not by itself measure relative causal impact on internal reasoning.

The annotation instrument has uneven validation. In a 27-trace expert-coded subset, agreement is high for `cruelty` and `personal_cost_burden` (both $\kappa = 1.00$ for causality coding) and moderate-to-substantial for `overcorrection_concern` ($\kappa = 0.64$), whereas several psychological categories have near-zero agreement (Table 7). Interpretations of subtle self-presentational categories must therefore remain tentative.

6.6. Consequence framing and reasoning in Hassner and Savary

The completed N-series annotation contains 840 traces, 480 for Hassner and 360 for Savary (20 per design cell). Unlike historical centrality coding, these categories have binary presence indicators. In Hassner, the N−–N0 increase in `incentive_awareness` is 18.75 percentage points (pp; 95% CI [11.74, 26.10]), and the N+–N0 increase is 22.50 pp ([15.17, 29.72]). These mentions reflect, at least partly, consequences explicitly stated in the instructions. The N−–N0 change in `emotional_reasoning` is -13.75 pp in GLM ([-24.59, -2.90]) but +21.25 pp in Kimi ([8.02, 32.94]).

In Savary, incentive_awareness increases by 23.33 pp under N+ relative to N0 ([14.93, 31.15]), although its cluster-based multiple-testing-adjusted result is not significant ($q = 0.558$). The N+–N0 contrast for possibility_of_different_opinions differs between models by 31.67 pp (GLM minus Kimi; 95% CI [18.54, 44.80], $q = 0.0142$). Within matched design cells, source_reliance is positively associated with donation willingness (normalized ordinal-rank difference +0.100, 95% CI [0.054, 0.146], BH $q = 0.00486$). These are associations, not mediation estimates. For Hassner incentive_awareness, only 10 of 66 occurrences have known argumentative direction; sign changes relative to N0 cannot be estimated when the category is absent at baseline.

6.7. Reasoning in Bolsen and Corbacho: prevalence and direction

The analyzed sample covers 1,398 eligible traces (799 Bolsen; 599 Corbacho), two models, all five system stimuli, and only four of Bolsen's nine original treatments. The new codebook records category presence as binary (0/1), not the historical three-level centrality scale; findings from the two annotation pipelines are therefore reported separately. Regression estimates adjust for original treatment, model, and annotation version.

Compared with N0, A and C markedly increase the prevalence of overcorrection_concern: by 50.7 and 51.3 percentage points (pp) in Bolsen, and 70.8 and 57.9 pp in Corbacho, respectively (all BH-adjusted $q \leq 0.006$). This category closely reflects the A/C instructions, so greater prevalence need not represent deeper or more human-like deliberation. Domain-specific differences are also visible: C increases economic_payback in Bolsen by 25.6 pp (95% CI $[12.2, 39.0]$, $q = 0.006$), whereas A increases corruption_normalization in Corbacho by 49.1 pp ($[36.6, 61.6]$; $q < 0.001$). The corresponding N−/N+ prevalence contrasts are generally smaller and less precise. There is pronounced model heterogeneity: the Bolsen A–N0 increase in overcorrection_concern is 2.6 pp for GLM and 98.7 pp for Kimi. These contrasts measure annotated textual features, not causal mechanisms.

Figure 3.

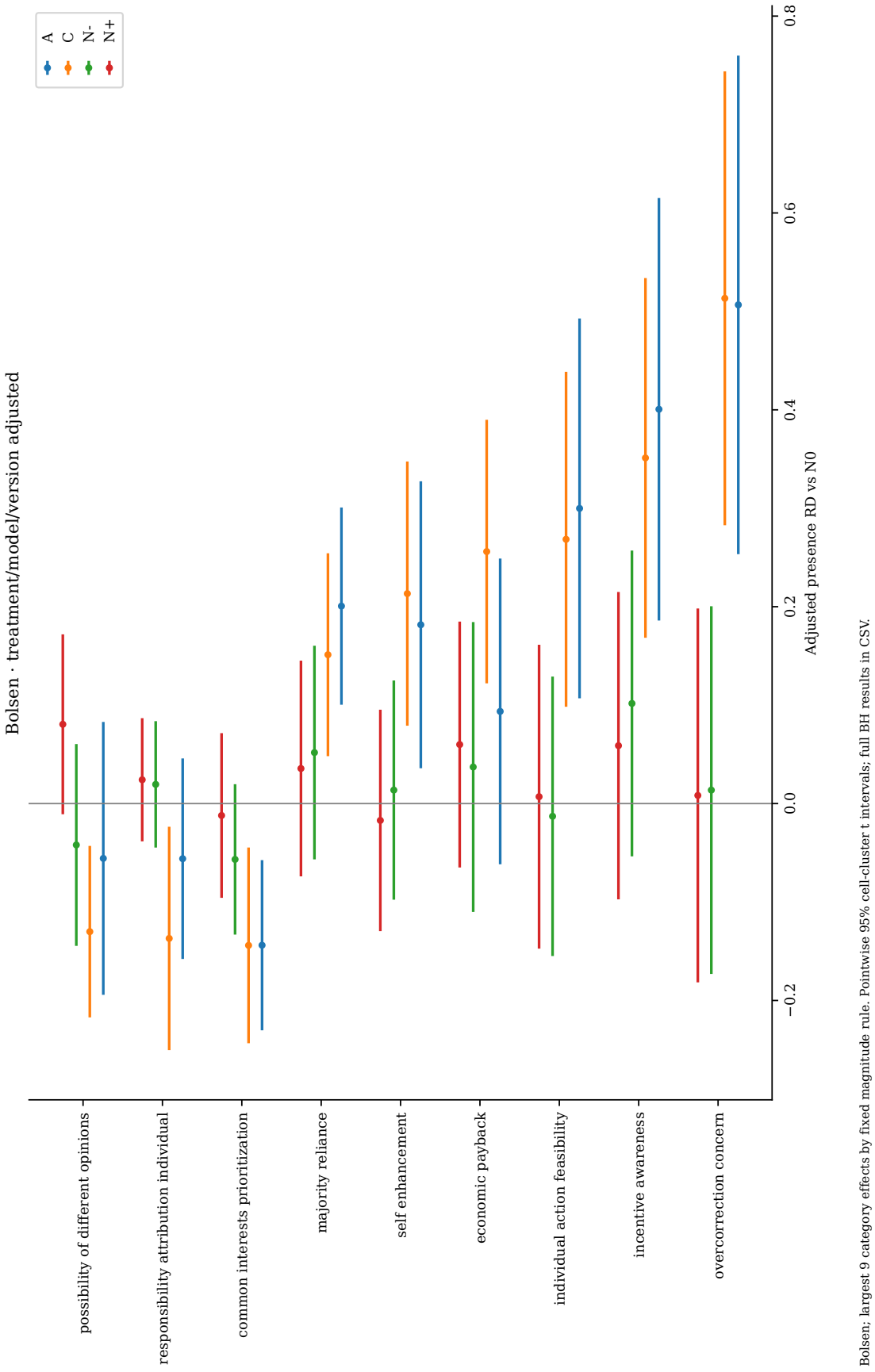


Figure 3: Bolsen category-presence differences from N0, adjusted for treatment, model and annotation version. Nine categories selected by effect magnitude; pointwise 95% cell-cluster t intervals, not simultaneous intervals. Only four of nine treatments are covered.

The direction analysis separates category occurrence from the sign of an argument among traces where it occurs. Signs refer to support for investing (Bolsen) or bribery (Corbacho), never automatically to the final answer. Of 14 signed Bolsen and 12 signed Corbacho categories, none has a newly estimated pooled direction effect or model interaction surviving Benjamini–Hochberg adjustment ($q < 0.05$). For example, in Bolsen A raises `majority_reliance` prevalence by 20.1 pp, while the conditional share of positively signed instances is estimated to fall by 35.6 pp (95% CI $[-70.3, -1.0]$; $q = 0.728$). This exploratory contrast illustrates why increased occurrence cannot be equated with more positive argumentative direction. Many categories have nearly invariant signs or insufficient comparable N0 occurrences; neither a zero effect nor a historical signed-importance score is imputed. Annotation validity, restricted treatment coverage, and conditioning on category presence further limit inference.

6.8. Associations between reasoning annotations and answers

Adding annotation features increases reported explanatory fit in Savary from $R^2 = 0.419$ to 0.482 ($\Delta R^2 = 0.063$). Overcorrection concern is negatively associated with willingness ($\beta = -0.047$, $p = 0.012$), while positively directed personal-cost and private-interest codes are associated with higher willingness (both reported $\beta = +0.046$). A gradient-boosting model also outperforms the linear benchmark in cross-validation (reported $R^2 = 0.558$ versus 0.475). These patterns indicate predictive information in verbalized reasoning beyond the specified treatment, stimulus, and model controls, but do not demonstrate that the generated text is causally upstream of the answer.

For Hassner, the original non-tautological annotation specification reports a smaller increment in fit ($\Delta R^2 = 0.052$). Positive directions of majority-reliance and means–ends arguments are associated with endorsement, while overcorrection concern itself is not a significant predictor ($p = 0.92$). Taken together, the two experiments suggest that the *observable association* between reasoning categories and final responses varies by domain. The small expert-validation sample and co-generation of traces and answers limit stronger process claims.

Associations in the new experiments. Within matched model–original-treatment–stimulus cells, the presence of `economic_payback` in Bolsen is

associated with a $+0.076$ difference on the normalized ordinal midrank outcome (95% CI $[0.040, 0.113]$, multiplicity-adjusted $q = 0.003$); a mean-score sensitivity estimate is $+0.204$ points ($[0.111, 0.297]$). These are associations between jointly generated text and answers, not evidence of mediation. In Corbacho, positive associations of some private-interest and means–ends categories with *True* do not survive multiplicity adjustment, and refusals remain a distinct nominal outcome. Low within-cell variation, uneven semantic validation, and the two annotation versions constrain these estimates.

6.9. Exploratory hidden-state results

The local Qwen3-8B analysis covers 98 runs (56 Hassner and 42 Savary) under a different stimulus inventory from the API study. In matched comparisons, the reported cumulative hidden-state trajectory is shorter under stimulus conditions than without a stimulus (3,370 versus 3,444 distance units; paired Wilcoxon $p < 0.001$). This is a geometric difference across layer representations, not a direct measure of computational effort or deliberation.

The between-/within-condition separation statistic peaks near layers 20–21 in the reported comparisons. Linear CKA remains above 0.94 across layers (reported L20 CKA $= 0.994$), indicating broadly similar representational geometry despite detectable condition differences. These summaries do not uniquely identify a translation mechanism or a semantic processing stage.

The local model also exhibits a representational–behavioral dissociation: stimulus-related hidden-state differences are detectable even where its Hassner responses are near-saturated (reported $P(\text{Yes}) \approx 0.998$). A treatment-status probe performs poorly under leave-one-group-out validation despite higher standard cross-validation performance, cautioning against scenario-specific leakage. For Savary, a condition-residualized ridge probe reaches a reported leave-one-out $R^2 = 0.665$ at layer 27. The distinct layer peaks are descriptive: they do not establish when a decision is causally formed. Small samples, high-dimensional activations, and the separate model architecture constrain generalization.

6.10. Synthesis

Three results are most robust as a descriptive account of the historical corpus. First, model responses vary substantially across model families and between

the two social domains; the Savary public-recognition contrast reverses the human benchmark. Second, system-prompt stimuli consistently accompany shifts away from the normatively approved response direction in the reported descriptive comparisons, while their apparent magnitude depends on prompt condition and model coverage. Third, stimuli are associated with changes in verbalized reasoning and with measurable hidden-state differences in a separate local model, but neither track alone identifies the mechanism producing final answers.

The historical findings support the study of system prompts as consequential experimental inputs, not as neutral wrappers. They do *not* establish that steering responses away from social approval makes LLMs more human-like, that the observed attribute associations isolate framing valence, or that verbalized and representational measures constitute a causal chain. The additional Bolsen and Corbacho results above reveal further model- and domain-specific heterogeneity.

7. Selected Results Tables

Table 2: New consequence-framing stimuli versus neutral realism N0: model-specific treatment-balanced differences. Bolsen values are mean points on the seven-point scale; Corbacho values are percentage-point differences in True response shares. No corresponding refusal changes were observed for these two stimuli.

Experiment	**Model**	**N− – N0**	**N+ – N0**
Bolsen	GLM-4.6	+0.038	−0.014
Bolsen	Kimi	+0.104	−0.011
Corbacho	GLM-4.6	0.0 pp	0.0 pp
Corbacho	Kimi	+3.3 pp	0.0 pp

Table 3: Selected Bolsen treatment contrasts under neutral realism (N0) versus published human contrasts. Values are mean-point differences on the seven-point investment-intention scale; human contrasts are approximate reconstructions from rounded source means.

Treatment contrast	**GLM-4.6**	**Kimi**	**Human**
Government attribution vs. control	-0.840	-1.800	≈ -0.370
Cost framing vs. control	-1.060	-0.600	≈ -0.360
Individual attribution + environment vs. control	$+0.020$	$+0.020$	$\approx +0.450$
Individual $\times$ environmental interaction	0.000	-0.180	$\approx +0.120$

Table 4: Corbacho: Kimi response fractions by treatment and system prompt. Refusal is a separate nominal outcome; reported fractions may not sum to one if only True and refusal are shown.

Stimulus	**Original treatment**	$P(\text{True})$	$P(\text{Refusal})$
A	Control	0.560	0.380
A	Corruption information	0.694	0.306
A	Institutional-inefficiency placebo	0.840	0.160
C	Control	0.100	0.440
C	Corruption information	0.300	0.220
C	Institutional-inefficiency placebo	0.447	0.234

Table 5: Mean response by stimulus condition ($T = 0.01$, all treatment groups pooled)

Stimulus	**Hassner**		**Savary**	
	P(Yes)	*n*	Mean	*n*
no_incentive	19.1%	2,023	3.74	1,240
C_balanced_realism	41.6%	1,340	3.44	1,034
H_science_personal_gain	56.4%	399	3.35	300
I_science_public_good	59.9%	399	3.30	300
J_company_personal_gain	61.4%	399	3.28	300
K_company_public_good	63.9%	399	3.30	299
A_original_core_realism	72.1%	696	3.32	515
L_science_public_good_positive	88.2%	400	3.20	300
M_company_personal_gain_positive	90.0%	399	3.14	298

Stimuli H–M were fielded on GLM-4.6 and Kimi only.

Table 6: Signed-presence means by stimulus condition (Hassner, $T = 0.01$, all models pooled). Positive = framed pro-endorsement; negative = framed anti-endorsement. † = validated ($\kappa \geq 0.44$); ‡ = not in validation sample (exploratory).

Category	**No stimulus**	**C**	**A**	Δ_{A-C}
ratio_of_goals_and_means‡	−0.24	−0.01	+0.42	$+0.43^{***}$
cruelty (direction)†	−0.67	−0.68	−0.34	$+0.34^{***}$
majority_reliance (direction)†	−0.25	+0.06	+0.41	$+0.35^{***}$
source_reliance‡	−0.42	−0.09	+0.16	$+0.24^{***}$

Table 7: Annotator validation against expert gold labels ($n = 27$ traces, Cohen's κ)

Category	κ	**Verdict**
cruelty (causality)	1.00	reliable
personal_cost_burden (causality)	1.00	reliable
personal_cost_burden (direction)	0.89	reliable
cruelty (direction)	0.85	reliable
private_interests_prioritization	0.82	reliable
public_recognition (causality)	0.76	reliable
public_recognition (direction)	0.65	reliable
treatment_awareness	0.73	reliable
legality_assessment	0.71	reliable
private_interests_prioritization (dir.)	0.67	reliable
overcorrection_concern	0.64	reliable
social_reward (direction)	0.52	moderate
treatment_effect_direction	0.46	moderate
emergency_urgency	0.44	moderate
privacy_factor	0.44	moderate
majority_reliance (direction)	0.44	moderate
emotional_reasoning	0.06	noise level
common_interests_ prioritization	0.00	noise level
incentive_awareness	0.00	noise level
self_deception	0.00	noise level
fear_negative_evaluation	-0.04	noise level
self_enhancement	-0.06	noise level

8. Discussion

Across four adapted vignette experiments, LLM treatment sensitivity varies by domain, model, and surrounding system instructions. Under the historical no-stimulus condition, Hassner treatment contrasts have the same direction as in the human study, whereas the Savary public-recognition contrast has the opposite sign. Bolsen shows partial directional correspondence but often different magnitudes. Corbacho produces uniformly negative answers under the neutral-realism reference N0; the corresponding human direct-response contrast is small and imprecisely estimated, limiting claims about replication failure.

The original A/C stimuli move Hassner and Savary answers away from the normatively approved response direction. This pattern is not general across the additional cases: relative to N0, A raises Bolsen energy-conservation willingness in GLM but lowers it in Kimi. In Corbacho, A/C change Kimi's affirmative and refusal shares substantially, and the institutional-inefficiency placebo elicits more affirmative answers than the corruption-information treatment. The N$-$/N$+$ shifts are generally modest on average, but Savary shows opposite directions in GLM and Kimi, and Hassner exhibits a substantial local control-cell change despite a small pooled shift. These patterns establish prompt sensitivity, not greater fidelity to humans or a uniform reduction in social desirability bias. N0 includes a realism instruction and is not equivalent to historical no stimulus.

Classical gain–loss framing effects therefore do not imply that consequence framing in a system prompt must shift synthetic answers. The generally small N$-$/N$+$ contrasts are compatible with context-dependent framing effects, but do not establish equivalence or prove a zero effect. Saturated N0 responses in Corbacho also limit observable shifts, while N0 itself contains a realism instruction. Historical H–M comparisons suggest a valence association, but their incomplete prompt factorial cannot isolate valence causally. In the new cases, A/C produce large changes in annotated category prevalence and strongly heterogeneous GLM/Kimi patterns. Bolsen payback arguments are associated with higher investment ratings within matched cells, whereas new direction contrasts do not survive multiplicity adjustment. Category presence, argumentative sign, and response association measure different features; none establishes mediation. Exploratory hidden-state differences in a separate local Qwen3-8B likewise

cannot explain the API models' responses. LLM simulations may help stress-test experimental wording, but comparisons against human treatment benchmarks remain necessary.

9. Scope and Limitations

The cases are not representative of human survey populations. Model coverage is unequal (historical H–M and new cases primarily GLM/Kimi), and low-temperature repetitions are not independent respondents. Corbacho's three nominal answers require separate treatment of refusal; its human privacy-adjusted estimate is not directly equivalent to our direct-response outcome. The N0 reference and historical no-stimulus baseline differ, as do some original vignette procedures. New `none` and N-series runs were executed sequentially with mixed API-provider routing, and 27 transport errors cluster in one GLM/Hassner/N− cell; neither cross-run wording effects nor local interactions are fully isolated.

The historical reasoning codebook was checked against only 27 expert-coded traces, with highly uneven agreement across categories. The new binary presence annotations lack equivalent human-gold validation, and the Bolsen annotations cover only selected original treatments. Reasoning-answer associations are predictive, not causal. Hidden-state measurements concern 98 runs of a different, local Qwen3-8B checkpoint; layer-wise geometry and probing are diagnostic rather than mechanistic proof. Historical H–M prompt wording remains unauthenticated verbatim; historical counts use a reconciled master-table copy.

10. Conclusion

System prompts materially affect synthetic experimental answers, but the direction and implications of those effects depend on task and model. Agreement with some human treatment contrasts coexists with reversals, attenuated effects, response saturation, and placebo sensitivity. Historical traces and separate-model representations provide complementary correlational evidence, not a validated causal pathway. Consequently, moving answers away from socially approved positions must not be equated with improved human realism. LLMs are potentially

useful for experimental stress-testing, but not as unvalidated substitutes for human respondents.

Declaration of the Use of Artificial Intelligence

AI was used for the following tasks:

- searching for articles for replication purposes (Consensus, Elicit AI, GPT-5.3 Instant), while all articles were independently read and reviewed by the authors;
- assisting in the selection of articles examining experimental designs in the context of large language models (Consensus, Elicit AI, GPT-5.3 Instant), while all articles were independently read and reviewed by the authors;
- assistance with translation into English and grammatical editing of the text written independently by the authors (Claude);
- assistance with data annotation based on a coding scheme and analytical categories manually developed by the authors (Claude Code);
- assistance with writing and debugging code for running the experiments (Claude Code);
- assistance with formatting the text according to LaTeX style requirements (ChatGPT 5.5).

References

Aher, G. V., Arriaga, R. I., & Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. *Proceedings of the 40th International Conference on Machine Learning (ICML)*, *202*, 337–371.

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, *31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2024). Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*, *32*(4), 401–416. https://doi.org/10.1017/pan.2024.5

Bolsen, T., Druckman, J. N., & Cook, F. L. (2014). Communication and collective actions: A survey experiment on motivating energy conservation in the u.s. *Journal of Experimental Political Science*, *1*(1), 24–38. https://doi.org/10.1017/xps.2014.2

Corbacho, A., Gingerich, D. W., Oliveros, V., & Ruiz-Vega, M. (2016). Corruption as a self-fulfilling prophecy: Evidence from a survey experiment in costa rica. *American Journal of Political Science*, *60*(4), 1077–1092. https://doi.org/10.1111/ajps.12244

Crowne, D. P., & Marlowe, D. (1960). A new scale of social desirability independent of psychopathology. *Journal of Consulting Psychology*, *24*(4), 349–354. https://doi.org/10.1037/h0047358

Cui, Z., Li, N., & Zhou, H. (2025). A large-scale replication of scenario-based experiments in psychology and management using large language models. *Nature Computational Science*, *5*(8), 627–634. https://doi.org/10.1038/s43588-025-00840-7

Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences*, *27*(7), 597–600. https://doi.org/10.1016/j.tics.2023.04.008

Hassner, R. E. (2023). Persuasive and unpersuasive critiques of torture. *Perspectives on Politics*, *21*(1), 160–173. https://doi.org/10.1017/S1537592721004138

Horiuchi, Y., Markovich, Z. D., & Yamamoto, T. (2022). Does conjoint analysis mitigate social desirability bias? *Political Analysis*, *30*(4), 535–549. https://doi.org/10.1017/pan.2021.30

Horton, J. J., Filippas, A., & Manning, B. S. (2023). *Large language models as simulated economic agents: What can we learn from homo silicus?* (NBER Working Paper No. 31122). National Bureau of Economic Research.

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). Large language models are zero-shot reasoners. *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 22199–22213.

Krumpal, I. (2013). Determinants of social desirability bias in sensitive surveys: A literature review. *Quality & Quantity*, *47*(4), 2025–2047. https://doi.org/10.1007/s11135-011-9640-9

Kühberger, A. (1998). The influence of framing on risky decisions: A meta-analysis. *Organizational Behavior and Human Decision Processes*, *75*(1), 23–55. https://doi.org/10.1006/obhd.1998.2781

Nederhof, A. J. (1985). Methods of coping with social desirability bias: A review. *European Journal of Social Psychology*, *15*(3), 263–280. https://doi.org/10.1002/ejsp.2420150303

O'Keefe, D. J., & Jensen, J. D. (2006). The advantages of compliance or the disadvantages of noncompliance? a meta-analytic review of the relative persuasive effectiveness of gain-framed and loss-framed messages. *Communication Yearbook*, *30*(1), 1–43. https://doi.org/10.1080/23808985.2006.11679054

Paulhus, D. L. (1984). Two-component models of socially desirable responding. *Journal of Personality and Social Psychology*, *46*(3), 598–609. https://doi.org/10.1037/0022-3514.46.3.598

Röttger, P., Hofmann, V., Pyatkin, V., Hinck, M., Kirk, H. R., Schütze, H., & Hovy, D. (2024). Political compass or spinning arrow? Towards more meaningful evaluations for values and opinions in large language models. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 15295–15311. https://doi.org/10.18653/v1/2024.acl-long.816

Salecha, A., Ireland, M. E., Subrahmanya, S., Sedoc, J., Ungar, L. H., & Eichstaedt, J. C. (2024). Large language models display human-like social desirability biases in Big Five personality surveys. *PNAS Nexus*, *3*(12), pgae533. https://doi.org/10.1093/pnasnexus/pgae533

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? *Proceedings of the 40th International Conference on Machine Learning (ICML)*, *202*, 29971–30004.

Savary, J., & Goldsmith, K. (2020). Unobserved altruism: How self-signaling motivations and social benefits shape willingness to donate. *Journal of Experimental Psychology: Applied*, *26*(3), 538–550. https://doi.org/10.1037/xap0000261

Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., Bowman, S. R., et al. (2024). Towards understanding sycophancy in language models. *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Tjuatja, L., Chen, V., Wu, T., Talwalkar, A., & Neubig, G. (2024). Do LLMs exhibit human-like response biases? a case study in survey design. *Transactions of the Association for Computational Linguistics*, *12*, 1011–1026. https://doi.org/10.1162/tacl_a_00685

Tourangeau, R., & Yan, T. (2007). Sensitive questions in surveys. *Psychological Bulletin*, *133*(5), 859–883. https://doi.org/10.1037/0033-2909.133.5.859

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 24824–24837.